

ISDN

Module 27

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hello viewers I welcome you to the lecture Series in cyber security and information security today we will discuss ISDN security ISDN that is integrated Services digital Network is a method to transfer voice data with some particular Services accessible for data ISDN is a set of CCI /it standards for circuit switch transmission of data over various media using ordinary telephone grade copper wire ISDN provide worldwide digital communication service in the transition to electronic documents and business transactions ISDN digital nature facilitates adding security but it has been deployed with a little thought to security ISDN provides digital circuit switched voice and data services as well as packet switch data services to users

ISDN is a method to transfer voice data with some particular Services accessible for data history of ISDN before the coming of age of ISDN original analogy telecommunication Services also known as plain old telephone system PS were the only source of telecommunication worldwide PS was the originally acronymic for post office telephone service system but changed when the post offices stopped offering telephone services PS comprised mainly of copper wires which link the subscribers home to the central switch office this form of telecommunication had its limitations longdistance calls had to be routed through operators and switchboards which made them highly unreliable and time consuming there was also the issue of static interference known as the line noise which Disturbed

communication in the 1960s the telecommunication industry began working on the converting its analog systems to transmit packets of digitized data by digital switching the United Nations International Telephone and Telegraph consultative committee now known as International telecommunication Union telecommunication iut actively pushed and encouraged the research for integrated Services digital Network by starting a movement to recommend and provide Sands for international digitization of the telecommunication services this was initiated in 1984 two major US networks Northern telecommunication and atnt took the first steps towards implementing ISDN but did not interoperate with existing equipments and software available to telecommunication networks this led to a setback for ISDN worldwide by 1990 due

to the worldwide effort National ISDN 1 ni1 was made compatible with existing proprietary equipments for telecommunication services this way people did not need to switch brand or buy software for the network they set the standards and procedures for future digital telecommunication technology to be used by everyone ISDN has resulted in better voice quality and internet access due to its packet switch connection with ISDN voice and data are carried by a bearer channel that is B Channel with a bandwidth of 64 kbps sometimes 56 kbps as opposed to telephone line which carry 52 kbps a data channel that is D channel is used for controlling Network Services and signaling to construct bearer connections and carry data over the bearer

channels this carries a bandwidth of 16 kbps or 64 kbps for now the future of ISDN lies with Broadband ISDN that is B-ISDN this entails transferring data voice and video all at the same time using fiber optic telephone line which can carry data rates from 155 Mbps to 622 Mbps and Beyond this is the major topic for research and development now we come to the types of ISDN there are two types of ISDN first basic rate ISDN that is B second primary rate ISDN that is p there are two services offered by ISDN this is known as the B and b1 b1rri basic rate interface consists of 2 kbps bearer channels and 11 16 kbps data Channel totaling 144 kbps as the name implies B usually has enough bandwidth for individual users PRI primary rate interface is used for clients that

require greater bandwidth it usually comprised of 2 364 kbps B channels and one 164 kbps D Channel totaling 1.536 Mbps sometimes the pr service can have 30 B channels and One D channel for a total of 1.984 Mbps b1rri has two Bearer that is B channels for transfer data which operate at 64 kbps each the other channel is known as the Delta D Channel and operates at 6 16 kbps the T channel is used for link management purposes and signaling this is known as 2B plus 1D PR depends on the country but a typical PR is 23 Bearer channels that is B channels at 64 kbps for data and One D Channel at 64 kbps one format of the pr interface consisted of 24 DS zeros and looked a lot like channelized T1 circuit the PRI is similarly defined for international use where the fundamental historical

transport bit rate is 2.48 Mbps sometimes called E1 in this case one of the 32 DS Z was already used for control purpose so there were only 31 available DS zeros for ISDN hence the format was 30b Plus D the major difference between a garden variety T1 and a PR is that the P normally has 23 channels used for transport of digital voice or data conversations called Bearer or B channels while the 24th channel was used for control called the data or D Channel thus a PRI ISDN circuits is often referred to as 23b plus d normally is inserted above because One D channel can control more than 23 B channels so it was also possible to have a 24b ISDN circuit so long as the control came from another PR so now we

come to the features of isdn the major feature of isdn is that it put together speech and information on the same line which were not presented in the classic telephone system ISDN services are ability to deliver maximum data within the combination of voice data video facts

over a single line while it provides at least two instantaneous connections ISDN makes available access to packet switch networks because it is a circuit switch telephone Network system therefore it provides better quality of voice data than an analog phone user can attach several devices to the line according to their need instead of purchasing many analog phone lines the ISDN B channel is responsible to provide greater data rate

so now we move on to the ISDN Services ISDN provides three types of services bearer Services T services supplementary Services Telly and supplementary Services represent the type of features and functions which are visible to end users while bearer Services represent the parts of the network which remain hidden from end users bearer Services facilitate the realtime communication of digital information between end users these Services mainly relate to network functions and account for OSI layers 1 to 3 an example of the bearer Services is that 64 kbps 8 KHz structured speed service this Services uses a data rate of 64 kbps together with 8 kHz timing information which structures the data into octet intervals for transmitting a pulse code modulated that is PCM speech signal the fact that

the signal represents speech is known to the network allowing it to employ Transformations which may not preserve bit Integrity but will result in good quality audio reproduction T services provide a set of higher level functions on the top of bearer services these services account for OSI layers 4 to 7 examples of teleservices are telephony Services which provide speech communication over a B Channel with control signaling over the D channel f similar Services which facilitate the communication of bitmap images over a B Channel with controlling signals over the D Channel tetch Services which facilitate the interchange and communication of textual as well as formatted documents over a B Channel with control signaling over the D

Channel supplementary Services enhance bearer and T services in an independent fashion examples of supplementary services are the Centre service emulates a private Network and provides a specialized feature to a set of subscribers The call transfer service allows a user to transfer an active call to a third party the call waiting service allows a user already engaged in a call to be informed of another incoming call the calling line ID service provides the calling parties address information to the called party although these Services all appear geared towards circuit switch telephone calls they are equally applicable to the packet switch data calls working of ISDN how does ISDN work one single transmission channel for communication is available in analog Network or

regular telephone lines that are provided to us by telephone companies therefore only one service can be carried at a time that is voice data video at same time whereas in ISDN lines there are the same pair telephone lines are logically divided into multiple channels they are normally divided into two channels B channel is the first type of Channel 64 kbps of data can be transferred using B Channel normally ISDN lines possess two B channels one channel is dedicated for voice and the other is dedicated for data communication the data transmission

takes place on one pair copper wire second type of channel is used for line and calling system setup is known as D Channel or Delta Channel also third

Channel on the other hand have 16 kbps of bandwidth concept of ISDN line working with the help of image is as shown ISDN protocols components and routers option te1 that is terminal equipment type one this is the ISDN telephone computer ISDN fax machine or whatever it is that you have hooked up to the ISDN phone line te2 that is terminal equipment type two this is the old analog telephone old style fax machine modem or whatever you use to hook up to the analog phone line it can also be other Communications equipment that is handled by a ta ta that is terminal adapter this lets old te2 stuff talk to the isda network it also adapts other kinds of equipment like ethernet interfaces to ISDN nt1 that is Network

termination type one this is the end of the line for the local phone company and the beginning of your house phone network nt2 that is Network termination type two in most homes this won't exist if you were a big company with your own private telephone system then this would be the guts of that telephone system LT that is line termination this is the physical connection of the telephone company ET exchange termination this is the local phone company's logical connection from your phones to the phone network letters r s Tu and V in the diagram are reference point that everyone uses to talk about each of these parts of the network the the r reference point is the interface between an old style telephone and

terminal adapter equipment since most homes won't have any N2 equipment the s and t reference points are usually one and the same and are sometimes called St bus different things happen in the different parts of the network different wiring requirements different data speeds different encoding Etc the reference Point V and LT and ET equipment are in the phone company's domain ISDN provides for digital transmission over ordinary telephone copper wire as well as over other media uses asdn uses circuit switching to establish a physical permanent Point to point connection from the source to the destination ISDN has standards defined by the itu that Encompass the OSI the bottom three layers of which are physical data link and network as shown

in the diagram difference between ADSL and ISDN there are some obvious differences between the two Services ISDN lines provide two voice channels or one 128 kbps data Channel whereas ADSL is only data line which is used for only data the part for ADSL is by a phone company through the medium of copper while which keeps the line stays alive and works even when local power fails whereas in ISDN lines power is required locally and does not work if the local power fails ISDN advantages the basic advantage of ISDN is to facilitate the user with multiple digital channels these channels can operate concurrently through the same one copper wire pair the digital signals broadcasting transversely the telephone lines ISDN

provides High data rate because of digital scheme which is 56 kbps BR ISDN using a channel aggregation protocol such as bonding or multi-link triple P supports an uncompressed data transfer speed of 128 kbps plus bandwidth for overhead and signaling PR transfers at 17 minutes, 1 second an even higher speed of up to 1 1920 kbps ISDN allows multiple devices to share a single line it is possible to combine many different Digital Data sources and have the information routed to the proper destination ISD and network lines are able to switch many fold devices on the single line such as faxes computers cash register credit card readers and many other devices these all devices can work together and directly be connected to the single line ISDN takes only 2 seconds to launch a connection while other modems take 30 to 60 second for establishment instead of the phone company sending a ring voltage signal to the ring the bell in your phone inband signal it sends a digital packet on a separate Channel that is out of band signal the outof band signal does not disturb established connections no bandwidth is taken from the data channels and call setup time is very fast as said earlier ISDN disadvantages there are some disadvantages associated with ISDN and they are as follows the disadvantage of ISDN lines is that it is very costly than the other typical telephone system ISDN line are more expensive when compared to the regular landline telephone

system ISDN provider companies and ISDN users are required to have special dedicated line and then can Encore at extra cost ISDN requires specialized Digital Services just like telephone company so now we come to the security issues which is more secure ISDN cable modem DSL T1 Etc on the physic IAL medium part EES dropping on a phone land line where dialup modems are used as physically easier than spying on the more modern mediums because the lower data rate more easily accommodates homemade Electronics of questionable quality also some of the protocols used over various mediums include encryption which may hinder spying attempts on the physical line alth the marrow adding in deserted Courtyards during Moon moonless nights crouching under Services boxes to

our tapping wires is a time honored spying tradition I believe that most attacks nowadays will be on a logical layer the attacker will try to hack into the ISP systems or the machines on the customer side be it a cable DSL modem or a not totally updated desktop systems and he will do so by sending IP packets from far away these are mostly orthogonal to the question of what medium is used logic attacks allow the attacker to try his luck on millions of targets from the comfort of his own basement instead of confronting rainy Gloom and ill-tempered cats on the random chance that the dozen or so homes that he will approach de night may contain something valuable electronically speaking and that no board Insomniac will spot him and call the police physical spying still exist

but is mostly employed on the specific targets the attacker does not want to spy on anybody he is after you personally at which point you should have more pressing needs proofing a some home network against a dedicated attacker is hard think about Tempest or even about looking at your display and or keyboard over your shoulder from the outside through a window a good

telescope a video camera with High FPS and you can record a typed password from about 200 M away so for the specifics of DSL T1 cable ISDN I would advise selecting the provider choose an ISP with a good reputé about security issues in particular with regards to whatever modem he provides that's more important security wise than the physical medium

and to abstract the whole question away never do sensitive stuff over the Internet unless you employ appropriate logical protection such as https so this was all about ISDN security thanks for watching this program thank you